



Ecological network analysis of an urban water metabolic system: Integrated metabolic processes of physical and virtual water

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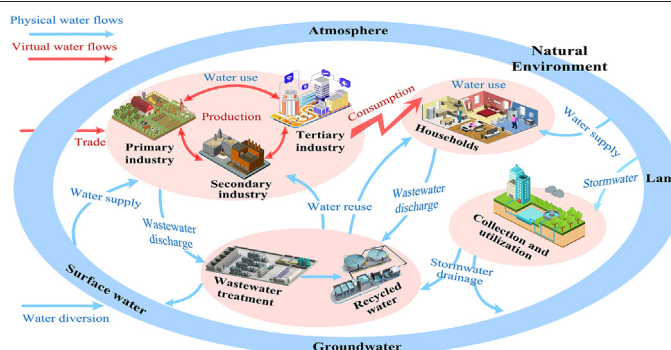
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HIGHLIGHTS

- An urban water flow structure including physical and virtual water is constructed.
- An ecological network model of an urban water metabolism system is developed.
- Main causes and critical compartments of unhealthy metabolic process are identified.
- The model and methodology are applied to promote local water management in Xining.

GRAPHICAL ABSTRACT



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ABSTRACT

The contradiction between social economy and water environment has become increasingly prominent, and the analysis of urban water metabolism system (UWMS) represents a problem-solving approach from the perspective of the entire flow process. However, a comprehensive UWMS model that considers both physical and virtual water flows is currently lacking. This paper presents an innovative application of an ecological network model of the UWMS-integrated metabolic process of physical and virtual water in Xining during the 2002–2018 period. By analysing and screening the metabolic characteristics, metabolic structure and metabolic relationships, the sustainability of the UWMS is evaluated in depth, and the main causes and critical compartments of the unhealthy metabolic process are identified. The findings show that the UWMS in Xining maintains a moderate level of robustness (the average $R = 0.48$) with limited metabolic efficiency. Since 2012, the water management policies in Xining have been significantly strengthened, contributing to a slight increase in robustness by improving the water use efficiency and metabolic structure. The integrated metabolic process is unhealthy because the metabolic structure is not reasonable, and the systematic metabolic relationship tends to be antagonistic due to the network mutualism index dropping to 1.0 during 2016–2018. We conclude that efficient irrigation management, more infrastructure projects for the sewer network, wastewater treatment and recycling could provide effective support to enhance the coordinate development of the social economy and water environment in Xining.

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1. Introduction

With the acceleration of socio-economic development and urbanization, water shortages and water pollution problems have become increasingly prominent (Eliasson, 2015; Zhang and Zuo, 2015; Pastori et al., 2017). Water-related environmental issues will undoubtedly

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hinder the sustainable and coordinate relationship between the social economy and water environment, especially in water-scarce regions and countries (Maqsood et al., 2005; Li et al., 2010; Huang et al., 2013; Cui et al., 2019). In order to relieve the anthropogenic pressure on natural water system and improve the efficiency of urban water utilization, it is crucial to promote the sustainable and coordinate development of both urban social economy and water environment, and identify the main causes for its unsustainability (Zheng et al., 2017; Zhai et al., 2019a; Zhai et al., 2019b). An analysis of urban water metabolism system (UWMS) is considered a good approach to solve such problems from the perspective of the entire flow process (Zhang et al., 2010; Paul et al., 2018).

The theoretical basis of urban water metabolism is derived from "urban metabolism", which was first proposed by Wolman in 1965 to support the sustainable development of a socio-economic system (Zhang et al., 2015; He et al., 2020). Urban water metabolism emphasizes the entire process of water quantity and water quality changes driven by socio-economic activities related to the development, utilization and protection of urban water resources (Wu et al., 2016; Zeng et al., 2016). A UWMS can effectively describe the structure and process of urban water flows, which is beneficial to deeper insight for sustainability studies reflecting the coordinate development of the social economy and water environment (Kharrazi et al., 2016; Zhang et al., 2019). A healthy metabolic process indicates that the system's compartments are able to support and coordinate with each other, reduce water consumption and alleviate the wastewater load of the natural system (Beloin-Saint-Pierre et al., 2017; He et al., 2020). Accordingly, most water-related environmental issues are associated with an unhealthy metabolic process, and the main causes and critical compartments need to be identified (Tambo, 2002; Yan and Wang, 2005; Zhang et al., 2010; Wu et al., 2016; Zeng et al., 2016).

The traditional metabolic process of a UWMS is mainly described by the physical water flows associated with the process of water supply, water use, water discharge, and water reuse (Zhang et al., 2010; Wu et al., 2016; Gao et al., 2019). However, under the background of economic globalization and trade regionalization, a large amount of physical water is exchanged throughout the process of the development and utilization of water resources, and a large amount of virtual water is embedded in commodities corresponding to the process of production, services, trade and consumption (Allan, 1998; Su et al., 2014; Zhao et al., 2015). Obviously, an integrated metabolic process combining physical and virtual water should be considered an important feature of a UWMS, because both physical and virtual water can interweave all compartments of socio-economic systems, and profoundly affect the circulation and redistribution of urban water resources (Fang et al., 2014; Hoekstra et al., 2016; Li et al., 2017; Gao et al., 2019). Therefore, it is necessary to pay more attention to the integrated metabolic process from the perspective of combining the two aspects, i.e., physical and virtual water.

Currently, ecological network analysis (ENA) is a main systematic analysis method that is initially used to identify the intrinsic attributes of an ecosystem, analyse the interactions among different compartments, and, thus, reveal the system's integrity and complexity (Fath, 2007). Given that the network structure of a UWMS also comprises multiple compartments linked by complex relationships, ENA is gradually applied in research investigating this system. Zhang et al. (2010) explored the physical water flow structure and metabolic relationships among different compartments in Beijing based on ENA. Wu et al. (2016) first used the method of grey water footprint to quantify the evolution of the water quality and then studied the metabolic structure and relationship in Dianchi Lake Basin based on ENA. Additionally, the virtual water flow structure and ecological relationship among various sectors in the ecological network of Guangdong were investigated by Zhang et al. (2019) based on an input-output model using ENA and by He et al. (2020) based on an optimized ecological hierarchy analysis (EHA) and network utility analysis (NUA). These studies are beneficial

to improve the current understanding of the metabolic structure and metabolic relationships of the UWMS, however, they are limited to the separate metabolic process of either physical water or virtual water.

In recent years, researchers have begun to pay more attention to the combination of physical and virtual water that generally transferred through water resource allocation and product trading. Zhao et al. (2015) first reported the quantification of China's physical and virtual water flows and associated water stress at the provincial level; Ye et al. (2018) developed a multi-objective optimization model to optimize the allocation of physical water resources integrated with virtual water trade to different water-demanding users in Beijing; Zhao et al. (2020) calculated virtual water imports expressed as food and energy trade and analysed the impact of physical and virtual water on urban water stress in Beijing. Whereas, these studies focused more on the transboundary flows and failed to form the entire process of water supply, water use, water discharge, and water reuse within the study area. Aiming at the deep insight of the entire metabolic process of a UWMS, the physical and virtual water transfer associated with a complete flow structure including both internal and external boundary processes is necessary, and needed to be researched deeply.

This article aims to conduct an in-depth evaluation of the sustainability of the UWMS, identify the bottlenecks restricting the sustainable utilization of urban water resources, and provide effective support to enhance the coordinate development of the social economy and water environment. We creatively develop an ecological network model of the UWMS-integrated metabolic process of physical and virtual water and apply this model to a typical water-scarce city, i.e., Xining during the 2002–2018 period (Section 2). To measure the sustainability of the UWMS, we focus on the system's integrity and analyse the long-term evolution of the metabolic characteristics based on indexes of metabolic efficiency, redundancy and robustness (Section 3.1). To further identify the main causes and critical compartments leading to the unhealthy metabolic process, we focus on the system's complexity and investigate the metabolic structure and metabolic relationships based on network structure analysis (NSA) and NUA (Sections 3.2 and 3.3). The main contributions of this study are as follows.

1) An urban water flow structure of the UWMS-integrated metabolic process of physical and virtual water is constructed; 2) an ecological network model of an UWMS is developed to reveal the system's integrity and complexity; 3) the main causes and critical compartments leading to unhealthy metabolic process are identified to promote local water management in Xining.

2. Materials and methods

2.1. Model construction

2.1.1. Flow structure of the UWMS-integrated metabolic process

As the fundamental substance supporting human life, physical water is allocated and utilized by humans in social and economic activities (Li et al., 2009; Kharrazi et al., 2016). Virtual water is transferred within or across regions along with trade and consumption during the economic process (Mao and Yang, 2012; Fang and Chen, 2015). The paths of physical water and the routes of trade or consumption of virtual water are interconnected, as reflected in the production and service process with physical water embedded into commodities and services to implement its conversion to virtual water (Hoekstra et al., 2011; Huang et al., 2013). Thus, the flow structure of a UWMS should be dominated by water flows involving both the physical field and virtual field.

Fig. 1 shows the flow structure of the UWMS-integrated metabolic process of physical and virtual water. The integrated metabolic process can be divided into three parts according to the different paths of physical and virtual water (Gao et al., 2019). The first part refers to the physical water flow process, which is defined as the development and utilization of natural water resources by the social and economic sectors, mainly including the process of water supply, water use, water

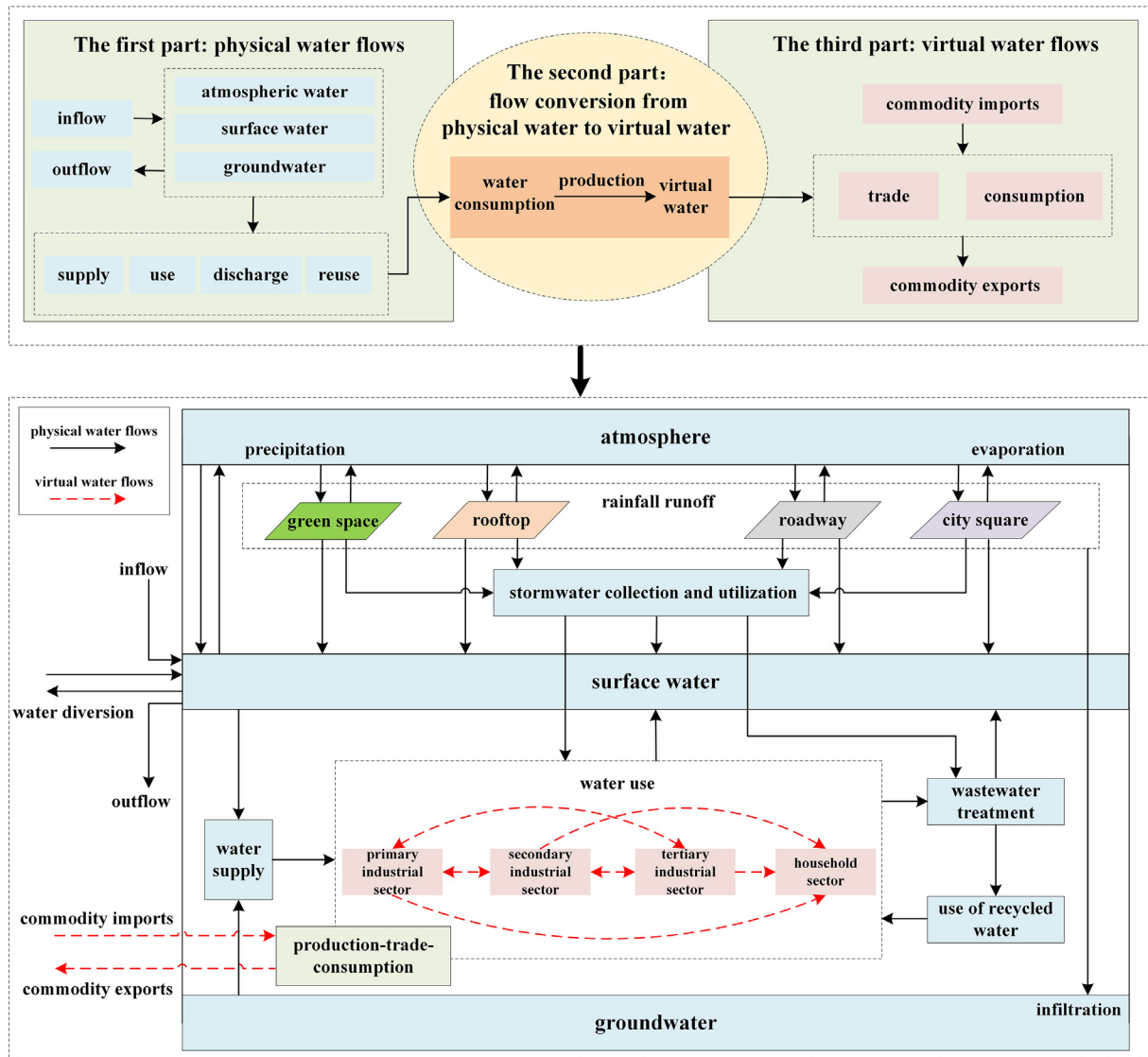


Fig. 1. Flow structure of the UWMS-integrated metabolic process of physical and virtual water.

discharge, and water reuse. The second part refers to the flow conversion process from physical water to virtual water. Physical water used in the process of production and service becomes virtual water embodied in corresponding commodities and services along the supply chain. “Consumption” refers to the loss of water from the available natural water bodies in a catchment area. Losses occur when water is incorporated into a product (Hoekstra et al., 2011). The third part refers to the virtual water flow process. In this part, people consume the virtual water embedded in commodities or transfer virtual water via commodity trade, leading to a change in the spatial and temporal distribution of virtual water.

2.1.2. Ecological network model of a UWMS

Fig. 2 shows the ecological network model of a UWMS. There are 8 compartments (Nat, SCU, Reu, Was, Hou, Ter, Sec and Pri) and 41 direct paths in the network. The compartments are the specific functional units that need to be identified first. To describe the physical water flows related to the natural water resources developed and utilized by the social and economic sectors, the compartment of the natural environment (Nat) is used to provide fresh water to meet the needs of urban users, including the compartments of household sector (Hou), primary industrial sector (Pri), secondary industrial sector (Sec), and tertiary industrial sector (Ter). Specifically, Pri includes agriculture,

forestry, animal husbandry and fishery. Sec includes mining, manufacturing, construction, and the supply of electricity, gas, water, etc. Ter includes all industries, except for primary and secondary industries, such as transport and storage, wholesale and retail trades, hotels and catering services, municipal service, information service, etc. (Huang et al., 2014; Liu et al., 2020). In addition, the virtual water flows associated with the process of production, services, trade and consumption are also connected to these water-using sectors. The compartment of wastewater treatment (Was) shows that the wastewater discharged from water-using sectors can be collected by urban wastewater treatment facilities. According to different treatment requirements, treated wastewater is discharged into the natural environment or urban water recycling facilities to obtain recycled water (Reu). The recycled water can be used for municipal green space irrigation, street cleaning, and industrial utilization after deep treatment. The compartment of stormwater collection and utilization (SCU) shows that some collected stormwater flows to wastewater treatment plants, while other stormwater flows to the natural environment.

The paths are the channels connecting the compartments by water flows. f_{ij} represents the flow originating from compartment j to compartment i , while z_j and y_j refer to the input flow and output flow between compartment j and the external environment, respectively. A steady-state network should meet the requirement that, for each

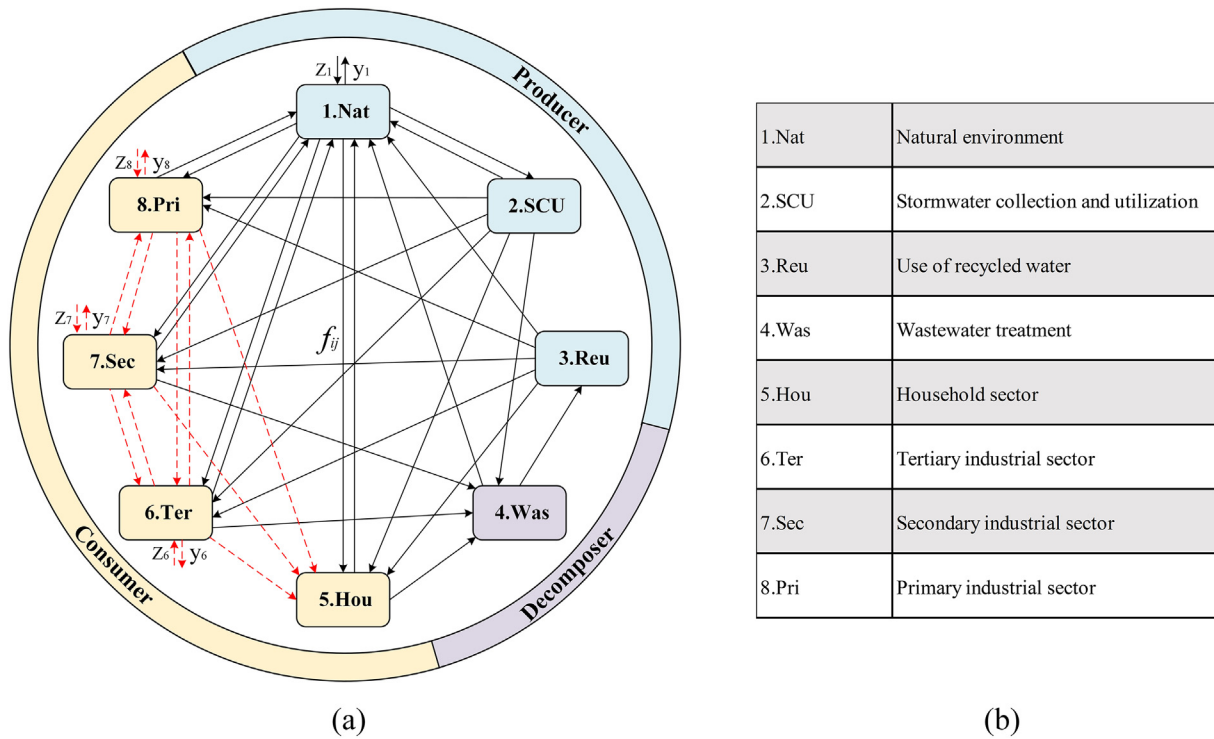


Fig. 2. Ecological network model of a UWMS. (a) Ecological network with physical water flows (shown by solid black arrows) and virtual water flows (shown by dotted red arrows); (b) Names and abbreviations of each compartment.

compartment, the total inputs equal the total outputs, including both inter-compartment flows and boundary flows, i.e., $T_j^{(in)} = T_j^{(out)}$ (Fath and Patten, 1999). Detailed descriptions of each path in the network are provided in the Supplementary Material (Table S1).

2.2. Information-based ENA

According to sustainability theory, a sustainable system should be able to exercise sufficient directional power to maintain its long-term integrity and have a flexible reserve response to external disturbances (Ulanowicz et al., 2009; Li and Yang, 2011). Thus, the system must maintain a balance between the following two equally important but complementary attributes: efficiency and flexibility (Ho and Ulanowicz, 2005; Kharrazi et al., 2013). Therefore, a UWMS requires a high level of robustness to support long-term sustainable development while maintaining integrity and stability, i.e., achieving a metabolic balance between metabolic efficiency and redundancy.

The information-based ENA can effectively evaluate the efficiency, flexibility, and more importantly, their trade-offs in a complex system by focusing on the transfer of flows between different compartments (Fang and Chen, 2015). Thus, the information-based ENA is used to investigate the metabolic characteristics of a UWMS, including its metabolic efficiency, redundancy and robustness.

2.2.1. Metabolic efficiency

Average mutual information (AMI) quantifies the amount of flow diversity encumbered by structural constraints (Rutledge et al., 1976; Ulanowicz and Wolff, 1991). Thus, AMI is used to estimate how flows are connected in an orderly and coherent manner and is selected as an index for the efficiency analysis. AMI is calculated as follows:

$$AMI = K \sum_{i,j} \left(\frac{f_{ij}}{T_{..}} \right) \log_2 \left(\frac{f_{ij} T_{..}}{T_i T_j} \right) \quad (1)$$

where $T_{..}$ represents the total flows among all compartments, T_j indicates the flows originating from compartment j , T_i denotes the flows to compartment i , and K is the scale coefficient.

In this study, the nonscaled AMI (i.e., $K = 1$) represents the authentic metabolic efficiency of the UWMS from only the structural perspective. AMI shows the capacity of the UWMS to perform in a sufficiently effective organization to ensure the network's long-term integrity. A higher AMI value indicates that the water flow structure of the UWMS is tightly constrained or highly organized, i.e., the flows in the network are more concentrated with fewer dispersion paths, corresponding to higher metabolic efficiency.

2.2.2. Redundancy

Conditional entropy (H_c) represents the amount of "choice" (residual diversity/freedom) pertaining to both the inputs and outputs of an average compartment in the network (Ho and Ulanowicz, 2005; Ulanowicz et al., 2009). Thus, H_c is used to gauge the disorder and freedom preserved by the network and is selected as an index for the redundancy analysis. H_c is calculated as follows:

$$H_c = -K \sum_{i,j} \left(\frac{f_{ij}}{T_{..}} \right) \log_2 \left(\frac{f_{ij}^2}{T_i T_j} \right) \quad (2)$$

In this study, the nonscaled H_c (i.e., $K = 1$) denotes the authentic redundancy of the UWMS only from a structural perspective. H_c shows the possibility of flows between relevant compartments due to the existence of some paths, reflecting the capacity of the UWMS to respond to disturbance by achieving a different stable state. A higher H_c value indicates that the water flow structure of the UWMS is less organized, i.e., more alternate paths exist in the network to contain unidentified flows, corresponding to higher redundancy. Appropriate redundancy plays a positive role in maintaining the stability of the UWMS and reducing its risk of collapse when facing internal or external changes.

However, an overly redundant network may become more disadvantageous for the system's process.

2.2.3. Robustness

A main strength of the information-based ENA approach lies in its ability to interpret the balance between efficiency and redundancy by using a single metric, i.e., robustness (R), to show the degree of constraint and degree of freedom in the network (Bodini et al., 2012). By concentrating on the roles of both efficiency and redundancy, R allows the creation of a generalized but quantitative measure of the sustainability of a complex network (Ulanowicz, 2009). R can be obtained by calculating an index of the order degree (a , i.e., relative efficiency) as follows (Goerner et al., 2009; Ulanowicz et al., 2014):

$$a = \frac{AMI}{AMI + H_c} \quad (3)$$

$$R = -a \log_2(a) \quad (4)$$

The relationship between a and R can be described by an inverted U-shaped curve. When $a < 0.37$, the UWMS is relatively redundant, which may lead to system stagnation with limited metabolic efficiency. When $a > 0.37$, the UWMS is relatively efficient, which may cause the system to be more vulnerable as excessive efficiency sacrifices redundancy. When a is close to 0.37, i.e., R approaches its peak value ($R > 0.5$), the UWMS attains a high level of robustness by reaching a metabolic balance between efficiency and redundancy, thus, the system supports long-term sustainable development while maintaining integrity and stability.

2.3. Network structure analysis

From the perspective of system complexity, a healthy or an unhealthy metabolic process is normally associated with the contribution of each compartment to the whole system, i.e., the metabolic structure of the UWMS. The weight W_j of each compartment in the system is calculated based on NSA. W_j reflects the influence and promotion of flows connected to one compartment on the integrated metabolic process. W_j is calculated as follows (Fath, 2004; Fath and Borrett, 2006):

$$G = (g_{ij}) = \frac{f_{ij}}{T_j} \quad (5)$$

$$N = (n_{ij}) = G^0 + G^1 + G^2 + G^3 + \dots + G^n = (I - G)^{-1} \quad (6)$$

$$\Lambda = (\lambda_{ij}) = N \text{diag}(T) \quad (7)$$

$$W_j = \sum_{i=1}^n \lambda_{ij} \left/ \sum_{i=1}^n \sum_{j=1}^n \lambda_{ij} \right. \quad (8)$$

where G represents the dimensionless direct flow matrix, $G^0 \sim G^n$ denote the flow intensities among the compartments that contain extended paths of 0 ~ n steps, I indicates the identity matrix, N denotes the dimensionless integral flow matrix, and Λ represents the dimensional integral flow matrix.

Taking the trophic functions of natural ecosystems as a reference, we define the compartments of the UWMS as producers, consumers and decomposers, which is helpful for further investigating the metabolic structure of the UWMS and the roles, statuses and functions of the key compartments. Specifically, Nat, SCU and Reu, which are related to the water supply, are defined as producers; Hou, Ter, Sec and Pri, which are related to water use, are defined as consumers; and Was, which is related to wastewater treatment, is defined as a decomposer.

2.4. Network utility analysis

In addition to the metabolic structure, a healthy or an unhealthy metabolic process may be affected by the mutualistic or antagonistic metabolic relationship of the UWMS. The integral utility matrix U and the corresponding symbol matrix $SignU$ are calculated to investigate the metabolic relationships among the compartments based on NUA. U is calculated as follows (Fath and Borrett, 2006; Fath, 2007):

$$D = (d_{ij}) = \frac{(f_{ij} - f_{ji})}{T_j} \quad (9)$$

$$U = (u_{ij}) = D^0 + D^1 + D^2 + D^3 + \dots + D^m = (I - D)^{-1} \quad (10)$$

where D denotes the direct utility matrix, $D^0 \sim D^m$ represents the flow utilities among the compartments that contain extended paths of 0 ~ m steps, and U indicates the dimensionless integral utility matrix.

There are four types of possible metabolic relationships among the compartments of a UWMS as follows: (+, -) denotes an exploitative relationship, (-, +) represents an exploited relationship, (+, +) indicates a mutualistic relationship, and (-, -) denotes a competitive relationship. From a whole system perspective, the network mutualism index (MI) and network synergism index (SI) are calculated to study the systematic metabolic relationship. MI refers to the ratio of the number of positive signs ($SignU(+)$) to the number of negative signs ($SignU(-)$). SI refers to the summation of all elements of U to assess the magnitude of the positive and negative relationships. MI and SI are calculated as follows (Fath, 2007; Lobanova et al., 2009):

$$MI = \frac{SignU(+)}{SignU(-)} \quad (11)$$

$$SI = \sum_{j=1}^n \sum_{i=1}^n u_{ij} \quad (12)$$

MI and SI can be combined to determine the systematic metabolic relationship, which tends to be mutualistic or antagonistic. If the positive relationship is dominant ($MI \geq 1$ and $SI \geq 0$), the systematic metabolic relationship is mutualistic; otherwise, the negative relationship is dominant ($MI < 1$ or $SI < 0$), and the systematic metabolic relationship is antagonistic.

2.5. Study area and data sources

Xining is the capital of Qinghai Province, and located in the Huangshui River Basin, which is a semi-arid area in northwest China (Fig. 3). The per capita water availability in Xining is 570 m³, accounting for only a quarter of that in China. Water resource scarcity has restricted local agricultural and socio-economic development over the long term (Bao and Zou, 2018). Meanwhile, the economy and trade in Xining have maintained steady growth as foreign trade increases from 1.33 billion to 3.13 billion from 2002 to 2018, which is expected to cause a fundamental restructuring of local virtual water flows (XMBS, 2003–2019).

In order to obtain the long-term evolution of the metabolic characteristics, metabolic structure and metabolic relationships, the ecological network model of the UWMS covers the data series of physical and virtual water flows during the 2002–2018 period. The physical water flows are derived from the “China Urban Construction Statistical Yearbook”, “Qinghai Water Resources Bulletin”, and “Xining Statistical Yearbook” (MOHURD, 2002–2018; QWRD, 2002–2018; XMBS, 2003–2019). The virtual water flows are calculated based on the input-output model. The input-output tables of Xining are extended from those of Qinghai Province, suggesting that their industrial structures are similar. Please refer to the Supplementary Material for details regarding the methods used to calculate the water flows (Table S1, Method S1 and Method S2) and data sources (Table S2).

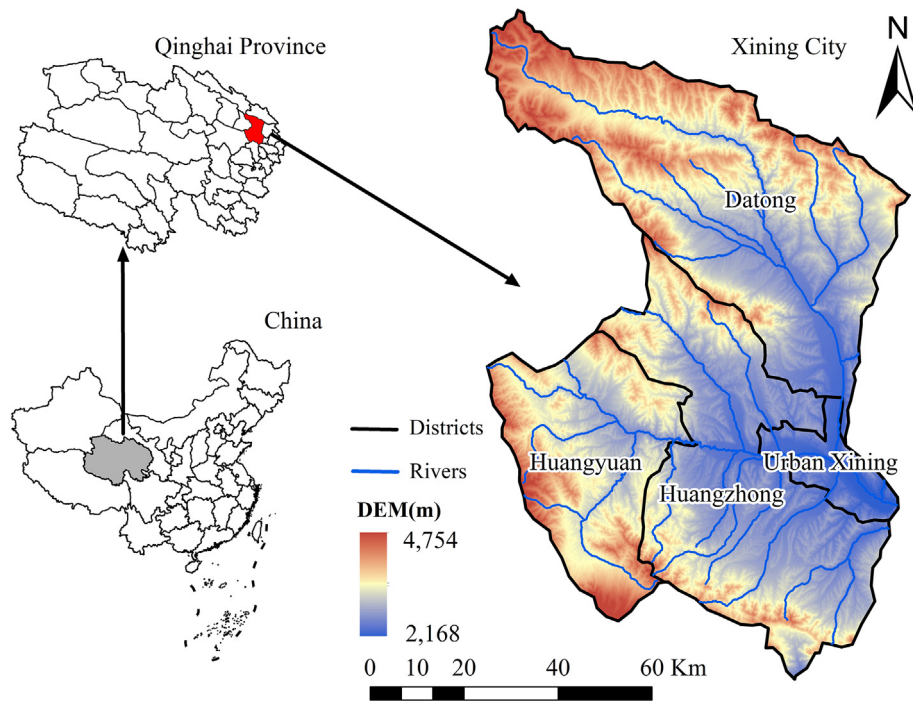


Fig. 3. Location of Xining in China.

3. Results and discussion

3.1. Metabolic characteristics of the UWMS in Xining

3.1.1. Evolution trends of AMI and H_c

The average AMI and H_c during the 2002–2018 period are 0.50 ± 0.04 and 1.84 ± 0.14 , respectively, and their variation trends differ

during the 2002–2011 period and become approximately similar since 2012 (Fig. 4a). AMI and H_c separately fluctuate at two different levels with average values of 0.50 ± 0.02 and 1.91 ± 0.05 during the 2002–2011 period, respectively. Then, AMI and H_c begin to follow the same downward trend and fluctuate at lower levels with average values of 0.45 ± 0.04 and 1.61 ± 0.13 during the 2012–2014 period, respectively. Finally, AMI and H_c rapidly grow and fluctuate at higher levels

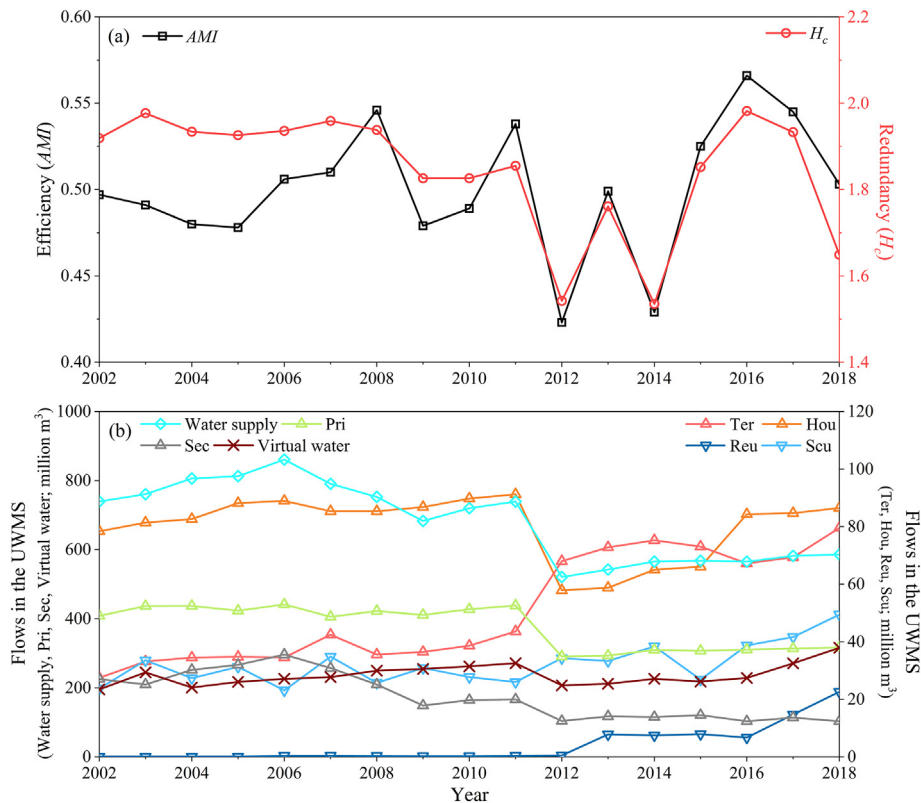


Fig. 4. Evolution trends of metabolic efficiency (AMI), redundancy (H_c) and main flows in the UWMS during the 2002–2018 period.

with average values of 0.53 ± 0.03 and 1.85 ± 0.15 during the 2015–2018 period, respectively. The variations in both AMI and H_c are related to the changes in the water flow structure. Fig. 4b shows the main flows in the UWMS associated with the water supply (Reu, Scu and local water sources), water use (Pri, Sec, Ter and Hou) and virtual water. Although these flows show a continuous and slow variation trend during most of the studied period, significant changes are observed at specific times, such as 2012 and 2016, further causing the variations in AMI and H_c .

External policy interventions have supposedly played an important role in causing the sudden changes in these flows. In 2012, the government of Xining has updated the regulations on the water supply and utilization management that promulgated in 2003 (Table 1). Thus, the flows associated with the water supply, water use (Pri, Sec and Hou) and virtual water tend to obviously decrease. Notably, the flow of water use (Ter) actually significantly increases at this time, which is caused by the drastic expansion of the tertiary industry. In addition, a series of strict water management policies associated with drinking water sources, water pollution, and agricultural water prices are significantly enforced in 2016 (Table 1). Thus, the flows associated with the water supply (Reu and Scu) and water use (Ter and Hou) change from a stable state to continuous growth after 2016. Although the variations in AMI (H_c) and the main flows in the UWMS are sensitive to external policy interventions, a simple linear relationship does not exist, because AMI (H_c) tends to decrease while the flows continue to show upward trends from 2017. This inference is confirmed by the linear regression analysis, which shows no significant correlation ($P > 0.01$) between AMI (H_c) and any single flow. As a result of the existence of indirect flows with complex nonlinear relations in the network, the current findings still have limitations in enhancing our understanding of the possible impact of different flows on metabolic efficiency and redundancy.

3.1.2. Relationship between a and R

The robustness curve shows that the values of R range from 0.46 to 0.49 during the 2002–2018 period (Fig. 5a). The average R is 0.48, indicating that the UWMS in Xining maintains a moderate level of robustness (0.4–0.5). Fig. 5a also shows that all 17 dots of R are scattered on the left side of the robustness curve, indicating that the UWMS is relatively redundant with the average a of 0.21. Consequently, the relatively high redundancy prevents the system from reaching a high level of robustness. In this case, although the high redundancy may play a positive role in maintaining stability, the water flow structure is not sufficiently organized due to some diversified and dispersed flows in the network, which may limit the metabolic efficiency and affect the system's long-term sustainable development.

Encouragingly, a has been gradually increasing during the study period, contributing to the increase in R from 0.46 in 2003 to 0.49 in 2018 (Fig. 5a and b). The evolution trends of a and R indicate that the UWMS strives to achieve a metabolic balance to obtain higher robustness. The increase in a during the 2006–2008 period is mainly due to the increase in AMI compared to the relative stability of H_c . The increase in AMI can be explained by the improvement in water use efficiency because the flows associated with the water supply and water use (Sec) began to

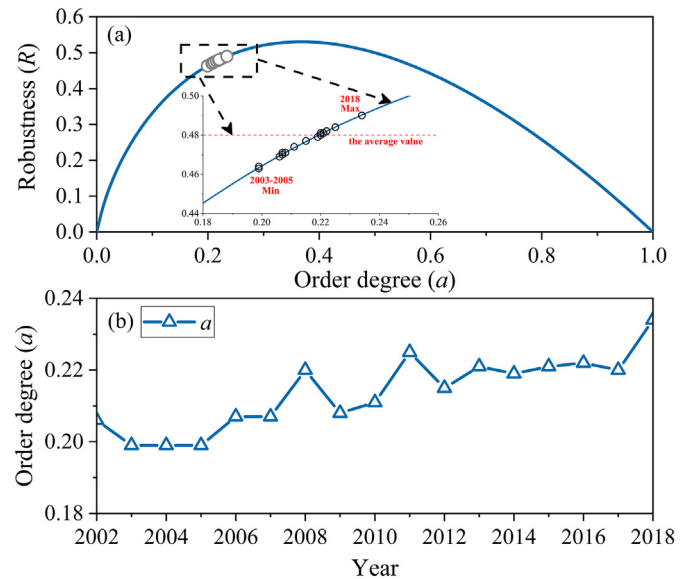


Fig. 5. Relationship between the order degree (a) and robustness (R) during the 2002–2018 period. (a) Robustness curve; (b) evolution trend of a .

decrease in 2006 (Fig. 4b). Additionally, the increase in a during the 2012–2018 period is likely related to the improvement in the water supply infrastructure. Since 2012, the flows associated with the water supply (Reu and Scu) rapidly increased, while the total water supply remained stable due to a series of strict water management policies contributing to saving water resources (Fig. 4b). The evolution trends of AMI , H_c and R confirm that Xining's water management policies are strengthened gradually and have a positive impact on improving the sustainability of the UWMS and obtaining a more effective water flow structure.

3.2. Metabolic structure of the UWMS in Xining

The evolution trends of W_j of the producers, consumers and decomposer from 2002 to 2018 are shown in Fig. 6a. The producers (Nat, SCU and Reu) contribute the most to the integrated metabolic process with an average W_j of $48.30 \pm 1.76\%$. The consumers (Hou, Ter, Sec and Pri) contribute slightly less than the producers with an average W_j of $43.59 \pm 2.15\%$. The decomposer (Was) contributes the least with an average W_j of $8.11 \pm 2.68\%$. Xining can basically reach a balance between the urban water supply and demand with a larger W_j of the producers than that of the consumers, especially during the 2012–2018 period. This finding also reflects that water management policies can effectively relieve the local water pressure and improve the metabolic structure, thus contributing to a slight increase in R .

However, the moderate level of robustness indicates that the integrated metabolic process of the UWMS in Xining is unhealthy to a certain extent. Thus, Section 3.2 aims to reveal the main causes and critical compartments of the unhealthy metabolic process related to the metabolic structure. Notably, the W_j of consumers is still quite considerable compared with the developed city (e.g. Beijing) with high water utilization efficiency, whose W_j usually remain below 40%; meanwhile, the decomposer contributes insufficiently compared with the developed city (e.g. Beijing) with advanced municipal construction, whose W_j are usually greater than 15% (Zhang et al., 2010). Therefore, although the metabolic structure of the UWMS in Xining can meet the basic requirement of the urban water demand, some improvement is still needed to obtain a healthy metabolic process.

The evolution trend of W_j of the producers remains relatively stable, but the related compartments Nat, SCU and Reu vary considerably (Fig. 6b). Nat accounts for the largest proportion of producers, suggesting that natural water bodies can serve as the most important sources of

Table 1
Important water management policies in Xining during the 2002–2018 period.

Policy list	Date
Regulations on water resource management in Xining	2003
Plan for water-saving society construction in Xining	2008
Regulations on water supply and utilization management in Xining	2012
Regulations for the protection and management of drinking water sources in Xining	2016
Action plan for water pollution prevention and control in Xining	2016
Implementation plan for the comprehensive reform of agricultural water prices in Xining	2017
Implementation plan for water-saving in Xining	2018

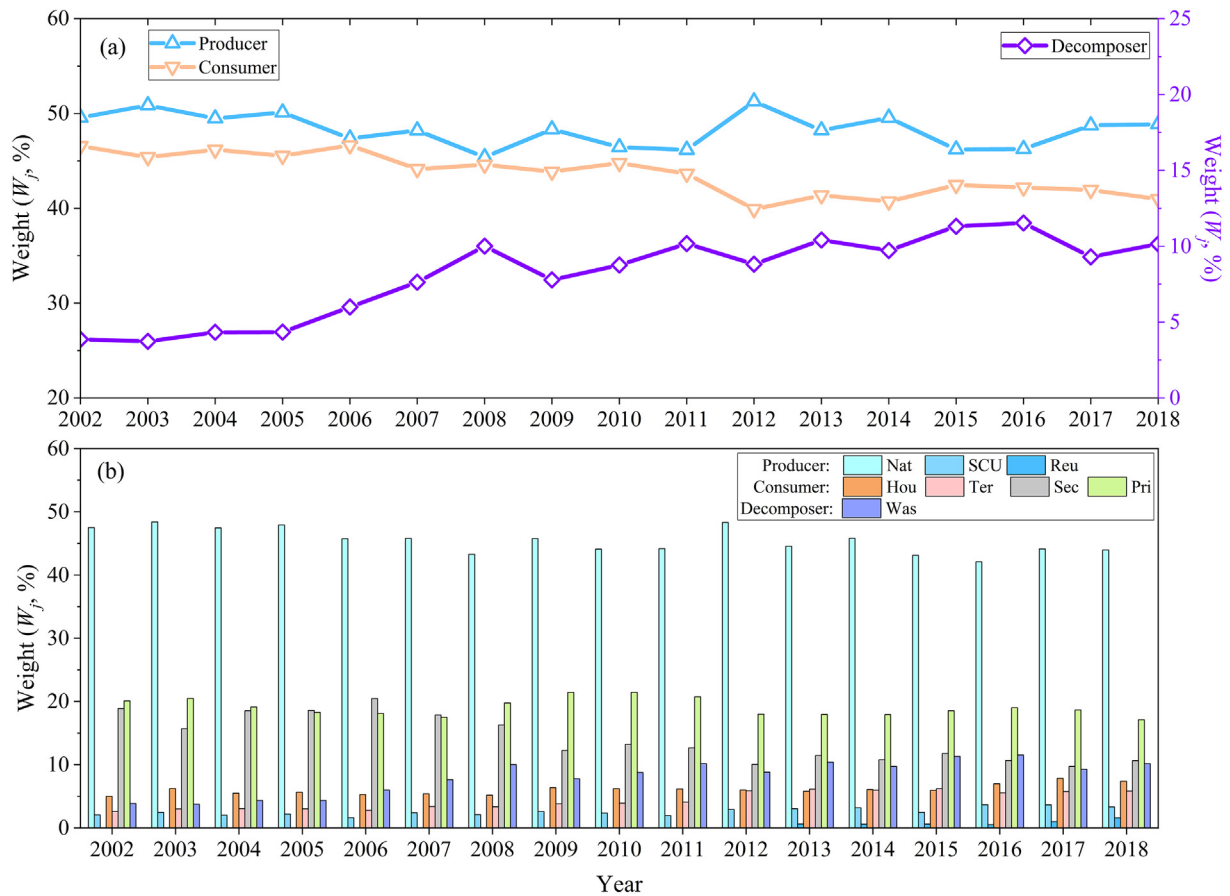


Fig. 6. Metabolic structure of the UWMS during the 2002–2018 period. (a) Evolution trends of the weight (W_j) of the producers, consumers and decomposer; (b) evolution trend of the weight (W_j) of each compartment.

the urban water supply. However, the W_j of Nat slowly decreases from 47.51% to 43.97% during the 2002–2018 period, which is likely caused by the slow reduction in local water resource reserves. In contrast, the W_j of SCU and Reu increases from 2.05% to 3.31% and from 0.01% to 1.59% from 2002 to 2018, respectively, indicating that Xining is vigorously developing the use of stormwater and recycled water to mitigate the decrease in natural water bodies. In particular, after 2016, the W_j of Reu significantly increases due to the construction and operation of several local recycled water plants with a daily water supply capacity of 75,000 tons. It is expected that the large-scale use of recycled water will significantly reduce the city's dependence on natural water bodies and improve the metabolic structure by increasing the contribution of producers.

The W_j of the consumers decreases from 46.56% to 40.98%, which should be mainly attributed to Sec, whose W_j decreases from 18.89% to 10.64% during the 2002–2018 period (Fig. 6b). This finding reflects that the secondary industry has made fruitful efforts in water conservation by promoting water-saving facilities and technologies. Pri begins to account for the largest proportion of consumers from 2008. The effective utilization coefficient of irrigation water in Xining is approximately 0.40, while China's average value of 0.55 (MWRPRC, 2018). Given the current low efficiency of agriculture water use associated with outdated technology and poor management, it is difficult to substantially reduce the contribution of Pri in the short term. Thus, Pri will continue to be a key factor limiting improvement in the metabolic structure. The W_j of Hou and Ter slightly increases because of the growth in the resident population (from 2,024,600 to 2,371,100) and the change in the industrial structure (proportion of the three industries varies from 7.3:42.7:50.0 to 3.6:36.4:60.0) during the 2002–2018 period in Xining (XMBS, 2003–2019).

The W_j of the decomposer (Was) increases from 3.87% to 10.15% from 2002 to 2018, suggesting that the wastewater treatment efficiency in Xining has steadily improved (Fig. 6b). If the W_j of Was continues to grow in the future, it could greatly benefit the improvement in the metabolic structure.

3.3. Metabolic relationships of the UWMS in Xining

Section 3.3 aims to reveal the main causes and critical compartments of the unhealthy metabolic process related to the metabolic relationships. Fig. 7 shows that the metabolic relationships of the UWMS in Xining can be divided into 6 different phases, namely, 2002–2005, 2006, 2007–2008, 2009–2011, 2012–2015 and 2016–2018. 28 pairwise relationships derived from 8 compartments show the same metabolic relationship during each specific phase. 9 pairwise relationships have changed during some phases, while the remaining 17 pairwise relationships remain unchanged throughout the study period. Notably, the blocks with a dark blue shadow in Fig. 7 indicate that the metabolic relationship of these two compartments differs from that during the previous phase.

The 9 changed pairwise relationships can be classified into three categories depending on their common features. The first category includes Ter-SCU, Sec-Ter, Hou-SCU and Ter-Nat. The changes in this category are relevant to their great competition for natural water bodies or limited wastewater treatment efficiency. For instance, Ter-SCU changes from (+, −) to (−, −) in 2012 because they are highly competitive for the limited wastewater treatment efficiency and some shared flow paths (i.e., the sewer network). Sec-Ter changes from (−, +) to (−, −) in 2006 because these two major water-using sectors are highly competitive for natural water bodies, the limited wastewater

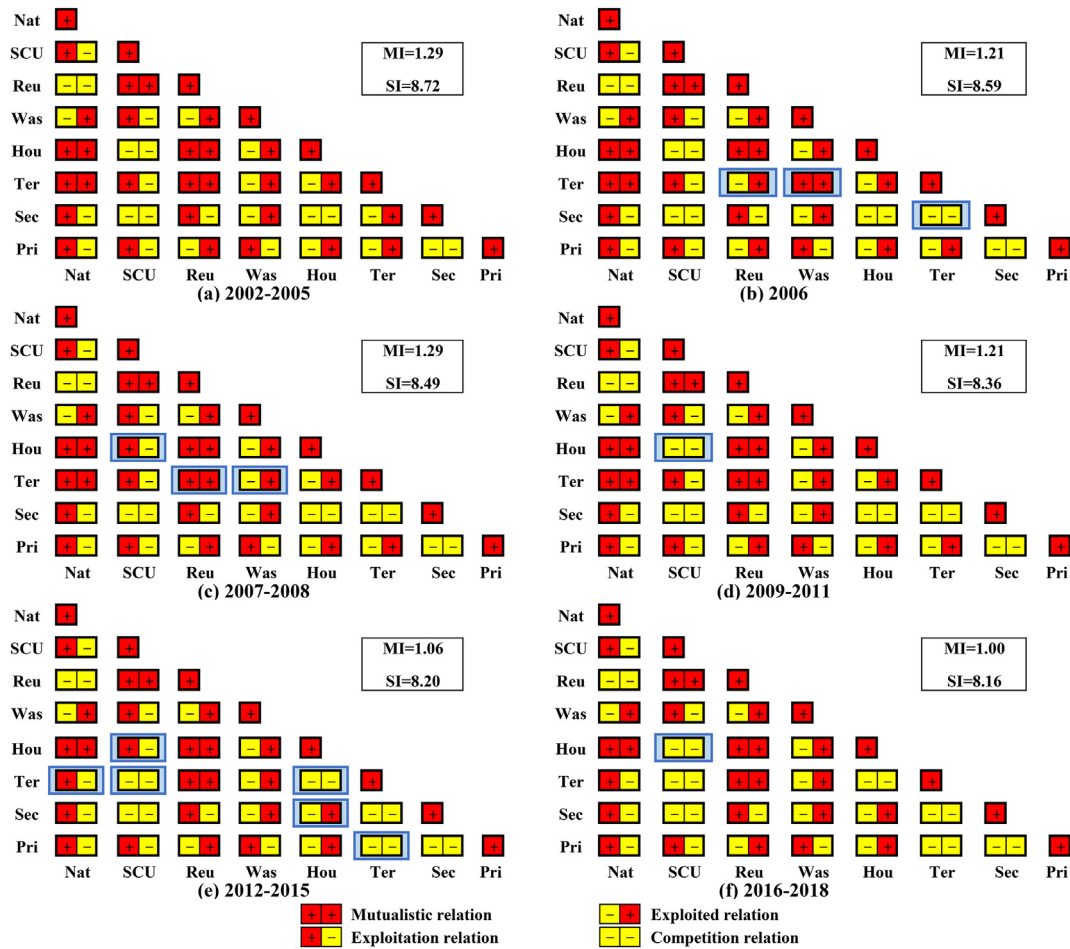


Fig. 7. Metabolic relationships among different compartments during the 2002–2018 period.

treatment efficiency and some shared flow paths. Hou-SCU shows a continuous change between $(-, -)$ and $(+, -)$ caused by the uncertainties in their water flows. When Hou-SCU shows $(-, -)$, they are highly competitive for the limited wastewater treatment efficiency and some shared flow paths; when Hou-SCU shows $(+, -)$, more stormwater can be reused for household living. Ter-Nat changes from $(+, +)$ to $(+, -)$ in 2012, which is highly relevant to the rapid development of the tertiary industry with an increasing amount of fresh water flowing from Nat to Ter.

The second category includes Ter-Hou, Pri-Ter and Sec-Hou. The changes in this category are mainly related to virtual water flows. For instance, Ter-Hou shows $(-, +)$ during the 2002–2011 period, which is probably due to the virtual water flow process, i.e., virtual water embedded in goods or services provided by the tertiary industrial sector can be transferred to the household sector via the path of consumption. Pri-Ter shows $(-, +)$ during the 2002–2011 period, which is probably caused by the virtual water embedded in agricultural products that are transferred to the tertiary industrial sector via the path of trade. Sec-Hou shows $(-, +)$ during the 2012–2018 period, which is probably caused by the increasing virtual water embedded in industrial products that are transferred to the household sector via the path of consumption.

The third category includes Ter-Reu and Ter-Was. The changes in this category only occur in one specific year. For instance, Ter-Reu shows $(+, +)$, except in 2006, during which it shows $(-, +)$, this is because the tertiary industrial sector can serve as both a consumer and a source of recycled water during most of the period. Ter-Was shows $(-, +)$, except in 2006, during which it shows $(+, +)$, this is because the tertiary industrial sector can discharge much more wastewater to wastewater treatment plants during most of the period.

The unchanged 17 pairwise relationships can be classified into four categories depending on four types of possible metabolic relationships. The first category includes 7 pairwise relationship with unchanged $(+, -)$. SCU-Nat, Sec-Nat and Pri-Nat show $(+, -)$, suggesting that a large amount of fresh water flows from Nat to SCU, Sec and Pri. Some collected stormwater flows to wastewater treatment plants, leading to $(+, -)$ of Was-SCU, while other stormwater flows to the natural environment for local irrigation, leading to $(+, -)$ of Pri-SCU. After treatment, most wastewater can be directly discharged into the natural environment for local irrigation, leading to $(+, -)$ of Pri-Was. Sec-Reu shows $(+, -)$, suggesting that the local secondary industrial sector can use recycled water on a large scale.

The second category includes 6 pairwise relationships with unchanged $(-, +)$. Hou-Was and Sec-Was show $(-, +)$, which is caused by the household sector and the secondary industrial sector discharging wastewater to wastewater treatment facilities. After treatment, wastewater flows to Nat and Reu, leading to $(-, +)$ of Was-Nat and Was-Reu. Pri-Reu shows $(-, +)$, further highlighting the negative role of the current backward irrigation technology and management, resulting in the low efficiency of recycled water. Pri-Hou shows $(-, +)$, which is probably caused by the consumption of agricultural products that drive the transfer of virtual water to the household sector.

The third category includes 3 pairwise relationships with unchanged $(+, +)$. Hou-Nat shows $(+, +)$, reflecting that the interactive flows between Hou and Nat are relatively balanced compared with Sec-Nat and Pri-Nat. Regarding Sec and Pri, the conversion of physical water to virtual water during the production process may cause an imbalance in their interactive flows. Hou-Reu shows $(+, +)$, suggesting that the household sector can serve as both a consumer and a source of recycled

water. Reu-SCU shows (+, +), reflecting that as water-saving measures, they can usually promote each other and develop a positive metabolic relationship.

The fourth category includes 3 pairwise relationships with unchanged (−, −). Reu-Nat shows (−, −), further proving that the low efficiency of recycled water can barely contribute to the positive relationship between recycled water and natural water bodies. Sec-SCU shows (−, −) because they are highly competitive for the limited wastewater treatment efficiency and some shared flow paths, even though some collected stormwater can be reused for industrial production. Pri-Sec shows (−, −) as both are highly competitive for natural water bodies.

The systematic metabolic relationship can be determined by the simultaneous analysis of *MI* and *SI*. We observe that *SI* maintains a higher value that is greater than 8.00 during the 2002–2018 period, while *MI* gradually decreases (Fig. 7). The values of *MI* are greater than 1.20 during the 2002–2011 period, decrease to 1.06 during the 2012–2015 period and further decrease to 1.00 during the 2016–2018 period. The systematic metabolic relationship of the UWMS in Xining is generally mutualistic with the positive relationship in the dominant position (i.e., $MI \geq 1$ and $SI \geq 0$) throughout the study period. However, *MI* decreases to a critical value of 1.00 during the 2016–2018 period, indicating that the systematic metabolic relationship shows a troubling trend towards antagonism. Therefore, the systematic metabolic relationship gradually becomes a potential downside that may lead to an unhealthy metabolic process of the UWMS in Xining.

The results of metabolic structure and metabolic relationships indicate that unhealthy metabolic process poses a serious challenge to the sustainability of the UWMS in Xining. First, it is difficult to significantly improve the metabolic structure due to limited efficiencies in irrigation, wastewater treatment and recycling, even if the secondary industry has achieved certain achievements in water-saving. Second, the systematic metabolic relationship is largely determined by three critical compartments of Pri, SCU and Reu. Great competitions for limited wastewater treatment efficiency and insufficient sewer network are mainly responsible for the negative relationship, while some water-saving measures, especially the large-scale use of recycled water, can always develop the positive metabolic relationship. Third, the virtual water flow process embedded among the water-using sectors (Pri-Hou, Sec-Hou, Ter-Hou, and Pri-Ter) is important to investigate the complexity of the UWMS.

4. Conclusions

This study establishes an ecological network model to reveal the integrity and complexity of the UWMS-integrated metabolic process of physical and virtual water. Through the analysis and screening of the metabolic characteristics, metabolic structure and metabolic relationships, the sustainability of the UWMS in Xining is evaluated in depth, and the main causes and critical compartments of the unhealthy metabolic process are identified. The model and methodology described in this paper provide deeper insights and applicable methods for sustainability study of a UWMS, which could be applied to any other water-scarce cities to promote local water management.

The UWMS in Xining maintains a moderate level of robustness with an average *R* of 0.48 during the 2002–2018 period. The system is relatively redundant due to the insufficient organization of the water flow structure, limiting the overall metabolic efficiency. However, the water management policies in Xining have been gradually strengthened, especially since 2012, contributing to a slight increase in robustness by improving the water use efficiency and metabolic structure.

The moderate level of robustness also indicates that the integrated metabolic process of the UWMS in Xining is unhealthy to a certain extent, because the metabolic structure is not reasonable, and the systematic metabolic relationship tends to be antagonistic. The metabolic structure and systematic metabolic relationship are mainly troubled by limited efficiencies of irrigation, wastewater treatment and recycling.

An accurate estimate of the systematic metabolic relationship could be achieved by considering the transfer of virtual water among the water-using sectors.

We believe that the future sustainability of the UWMS in Xining is highly dependent on efficient irrigation management and benefit from more infrastructure projects for the sewer network, wastewater treatment and recycling. However, due to some statistical restrictions and complex nonlinear relations in the network, the current findings still have some limitations in enhancing our understanding of water transfer characteristics, and the effectiveness of water management policies on metabolic efficiency, redundancy, and robustness. In future research, to depict clearly how water flow among the detailed compartments, the ecological network model should be further refined with more water-using sectors relating to Pri, Sec and Ter. Moreover, a multi-scenario simulation research based on the UWMS model will be conducive to investigating the possible effects of different water flows or specific policy recommendations on the integrated metabolic process.

CRedit authorship contribution statement

Dan Cui: Conceptualization, Data curation, Methodology, Data processing, Writing, Visualization.

Bingran Ma: Data collection, Visualization, Review.

Yue Zhuo: Data collection, Constructive discussions.

Yuxi Xie: Review, Constructive discussions.

Weihua Zeng: Review, Supervision.

Declaration of competing interest

We declare that we have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.scitotenv.2021.147432>.

References

- Allan, J.A., 1998. Virtual water: a strategic resource global solutions to regional deficits. *Ground Water* 36 (4), 545–546.
- Bao, C., Zou, J.J., 2018. Analysis of spatiotemporal changes of the human-water relationship using water resources constraint intensity index in Northwest China. *Ecol. Indic.* 84, 119–129.
- Beloin-Saint-Pierre, D., Rugani, B., Lasvaux, S., Mailhac, A., Popovici, E., Sibiude, G., Benetto, E., Schiopu, N., 2017. A review of urban metabolism studies to identify key methodological choices for future harmonization and implementation. *J. Clean. Prod.* 163, S223–S240.
- Bodini, A., Bondavalli, C., Allesina, S., 2012. Cities as ecosystems: growth, development and implications for sustainability. *Ecol. Model.* 245 (Complete), 185–198.
- Cui, D., Chen, X., Xue, Y.L., Li, R., Zeng, W.H., 2019. An integrated approach to investigate the relationship of coupling coordination between social economy and water environment on urban scale - a case study of Kunming. *J. Environ. Manag.* 234, 189–199.
- Eliasson, J., 2015. The rising pressure of global water shortages. *Nature* 517 (7532), 6.
- Fang, D.L., Chen, B., 2015. Ecological network analysis for a virtual water network. *Environ. Sci. Technol.* 49 (11), 6722–6730.
- Fang, D.L., Fath, B.D., Chen, B., Scharler, U.M., 2014. Network environ analysis for socio-economic water system. *Ecol. Indic.* 47 (dec), 80–88.
- Fath, B.D., 2004. Network analysis in perspective: comments on "WAND: an ecological network analysis user-friendly tool". *Environ. Model. Softw.* 19 (4), 341–343.
- Fath, B.D., 2007. Network mutualism: positive community-level relations in ecosystems. *Ecol. Model.* 208 (1), 56–67.
- Fath, B.D., Borrett, S.R., 2006. A MATLAB (R) function for network environ analysis. *Environ. Model. Softw.* 21 (3), 375–405.

- Fath, B.D., Patten, B.C., 1999. Review of the foundations of network environ analysis. *Ecosystems* 2 (2), 167–179.
- Gao, X.R., Sun, M., Zhao, Y., Wu, P.T., Jiang, S., Zhuo, L., 2019. The cognitive framework of the interaction between the physical and virtual water and the strategies for sustainable coupling management. *Sustainability* 11 (9), 2567.
- Goerner, S.J., Lietaer, B., Ulanowicz, R.E., 2009. Quantifying economic sustainability: implications for free-enterprise theory, policy and practice. *Ecol. Econ.* 69 (1), 76–81.
- He, C.Y., Huang, G.H., Liu, L.R., Xu, X.L., Li, Y.P., 2020. Evolution of virtual water metabolic network in developing regions: a case study of Guangdong province. *Ecol. Indic.* 108, 105750.
- Ho, M.W., Ulanowicz, R., 2005. Sustainable systems as organisms? *Biosystems* 82 (1), 39–51.
- Hoekstra, A.Y., Chapagain, A.K., Aldaya, M.M., Mekonnen, M.M., 2011. *The Water Footprint Assessment Manual: Setting the Global Standard*. Earthscan, London, UK.
- Hoekstra, A.Y., Chapagain, A.K., Zhang, G., 2016. Water footprints and sustainable water allocation. *Sustainability* 8 (1).
- Huang, C.L., Vause, J., Ma, H.W., Yu, C.P., 2013. Urban water metabolism efficiency assessment: integrated analysis of available and virtual water. *Sci. Total Environ.* 452, 19–27.
- Huang, Y.H., Dong, J., Zhuang, D.F., Zhu, Y.Q., Fu, Z.J., 2014. An improved approach for modeling spatial distribution of water use profit—a case study in Tuhai Majia Basin, China. *Ecol. Indic.* 36, 94–99.
- Kharrazi, A., Rovenskaya, E., Fath, B.D., Yarime, M., Kraines, S., 2013. Quantifying the sustainability of economic resource networks: an ecological information-based approach. *Ecol. Econ.* 90, 177–186.
- Kharrazi, A., Akiyama, T., Yu, Y.D., Li, J., 2016. Evaluating the evolution of the Heihe River basin using the ecological network analysis: efficiency, resilience, and implications for water resource management policy. *Sci. Total Environ.* 572, 688–696.
- Li, Y., Yang, Z.F., 2011. Quantifying the sustainability of water use systems: calculating the balance between network efficiency and resilience. *Ecol. Model.* 222 (10), 1771–1780.
- Li, Y., Chen, B., Yang, Z.F., 2009. Ecological network analysis for water use systems—a case study of the Yellow River Basin. *Ecol. Model.* 220 (22), 3163–3173.
- Li, Y.P., Huang, G.H., Nie, S.L., 2010. Planning water resources management systems using a fuzzy-boundary interval-stochastic programming method. *Adv. Water Resour.* 33 (9), 1105–1117.
- Li, Y.P., Nie, S., Huang, C.Z., McBean, E.A., Fan, Y.R., Huang, G.H., 2017. An integrated risk analysis method for planning water resource systems to support sustainable development of an arid region. *J. Environ. Inf.* 29 (1), 1–15.
- Liu, K.D., Yang, G., Yang, D.G., 2020. Industrial water-use efficiency in China: regional heterogeneity and incentives identification. *J. Clean. Prod.* 258, 120828.
- Lobanova, G., Fath, B.D., Rovenskaya, E., 2009. Exploring simple structural configurations for optimal network mutualism. *Commun. Nonlinear Sci. Numer. Simul.* 14 (4), 1461–1485.
- Mao, X.F., Yang, Z.F., 2012. Ecological network analysis for virtual water trade system: a case study for the Baiyangdian Basin in northern China. *Ecol. Inform.* 10, 17–24.
- Maqsood, I., Huang, G.H., Yeomans, J.S., 2005. An interval-parameter fuzzy two-stage stochastic program for water resources management under uncertainty. *Eur. J. Oper. Res.* 167 (1), 208–225.
- MOHURD, 2002–2018. *China Urban Construction Statistical Yearbook (2003–2019)*. Ministry of Housing and Urban-Rural Development, Beijing, China <http://www.mohurd.gov.cn/xytj/tjzljxxytjgb/jstjnj/>.
- MWRPRC, 2018. *China Water Resources Bulletin (2018)*. Ministry of Water Resources of the People's Republic of China, Beijing, China http://www.mwr.gov.cn/sj/tjgb/szygb/201907/t20190712_1349118.html.
- Pastori, M., Udias, A., Bouraoui, F., Bidoglio, G., 2017. A multi-objective approach to evaluate the economic and environmental impacts of alternative water and nutrient management strategies in Africa. *J. Environ. Inf.* 29 (1), 16–28.
- Paul, R., Kenway, S., McIntosh, B., Mukheibir, P., 2018. Urban metabolism of Bangalore City: a water mass balance analysis. *J. Ind. Ecol.* 22 (6), 1413–1424.
- QWRD, 2002–2018. *Qinghai Water Resources Bulletin (2002–2018)*. Qinghai Provincial Water Resources Department, Xining, China <http://slt.qinghai.gov.cn/>.
- Rutledge, R.W., Basore, B.L., Mulholland, R.J., 1976. Ecological stability: an information theory viewpoint. *J. Theor. Biol.* 57 (2), 355–371.
- Su, X., Li, J., Singh, V.P., 2014. Optimal allocation of agricultural water resources based on virtual water subdivision in Shiyang River basin. *Water Resour. Manag.* 28 (8), 2243–2257.
- Tambo, N., 2002. Hydrological cycle and urban metabolic system of water. *Water Waste-water Eng.* 28 (6), 1–5.
- Ulanowicz, 2009. The dual nature of ecosystem dynamics. *Ecol. Model.* 220 (16), 1886–1892.
- Ulanowicz, R.E., Wolff, W.F., 1991. Ecosystem flow networks - loaded dice. *Math. Biosci.* 103 (1), 45–68.
- Ulanowicz, R.E., Goerner, S.J., Lietaer, B., Gomez, R., 2009. Quantifying sustainability: resilience, efficiency and the return of information theory. *Ecol. Complex.* 6 (1), 27–36.
- Ulanowicz, R.E., Holt, R.D., Barfield, M., 2014. Limits on ecosystem trophic complexity: insights from ecological network analysis. *Ecol. Lett.* 17 (2), 127–136.
- Wu, B., Zeng, W.H., Chen, H.H., Zhao, Y., 2016. Grey water footprint combined with ecological network analysis for assessing regional water quality metabolism. *J. Clean. Prod.* 112, 3138–3151.
- XMBS, 2003–2019. *Xining Statistical Yearbook (2003–2019)*. Xining Municipal Bureau of Statistics, Xining, China <http://www.shujuku.org/statistical-yearbook-of-xining.html>.
- Yan, J.S., Wang, M.Z., 2005. The problems of water environment in the urban and their eco-essence. *Mod. Urban Res.* 4, 7–10. (in Chinese).
- Ye, Q.L., Li, Y., Zhuo, L., Zhang, W.L., Xiong, W., Wang, C., Wang, P.F., 2018. Optimal allocation of physical water resources integrated with virtual water trade in water scarce regions: a case study for Beijing, China. *Water Res.* 129, 264–276.
- Zeng, W.H., Wu, B., Chai, Y., 2016. Dynamic simulation of urban water metabolism under water environmental carrying capacity restrictions. *Front. Environ. Sci. Eng.* 10 (1), 114–128.
- Zhai, M.Y., Huang, G.H., Liu, L.R., Xu, X.L., Li, J.Z., 2019a. Transfer of virtual water embodied in food: a new perspective. *Sci. Total Environ.* 659, 872–883.
- Zhai, M.Y., Huang, G.H., Liu, L.R., Zheng, B.Y., Guan, Y.R., 2019b. Network analysis of different types of food flows: establishing the interaction between food flows and economic flows. *Resour. Conserv. Recycl.* 143, 143–153.
- Zhang, G., Huang, G.H., Liu, L.R., Niu, G., Li, J.Z., McBean, E., 2019. Ecological network analysis of an urban water metabolic system based on input-output model: a case study of Guangdong, China. *Sci. Total Environ.* 670, 369–378.
- Zhang, X.Y., Zuo, Q.T., 2015. Analysis of water resource situation of the Tarim River Basin and the system evolution under the changing environment. *J. Coast. Res.* 73, 9–16.
- Zhang, Y., Yang, Z.F., Fath, B.D., 2010. Ecological network analysis of an urban water metabolic system: model development, and a case study for Beijing. *Sci. Total Environ.* 408 (20), 4702–4711.
- Zhang, Y., Yang, Z.F., Yu, X.Y., 2015. Urban metabolism: a review of current knowledge and directions for future study. *Environ. Sci. Technol.* 49 (19), 11247–11263.
- Zhao, X., Liu, J.G., Liu, Q.Y., Tillotson, M.R., Guan, D.B., Hubacek, K., 2015. Physical and virtual water transfers for regional water stress alleviation in China. *Proc. Natl. Acad. Sci. U. S. A.* 112 (4), 1031–1035.
- Zhao, Y., He, G.H., Wang, J.H., Gao, X.R., Li, H.H., Zhu, Y.N., Jiang, S., 2020. Water stress assessment integrated with virtual water trade and physical transfer water: a case study of Beijing, China. *Sci. Total Environ.* 708, 134578.
- Zheng, M.S., Ma, L.K., Li, T.H., Huang, R., 2017. Rapid assessment method for eco-water deficit in catchments. *J. Environ. Inf.* 30 (2), 149–160.